Clinical Research Collaboration App

Product concept & functional documentation | Working draft — project name not finalized

1. Overview

This mobile application is intended to guide a research idea from early exploration to a reviewed research manuscript, while connecting doctors and researchers through a shared, AI-assisted workspace. The product has two role-based portals: a Researcher Portal and a Doctor / Research Specialist Portal. Both roles work on the same project record, with controlled access, discussion history, document versions, and clear stage progression.

Purpose

Help a researcher independently shape an initial idea with integrated AI support, then invite a suitable doctor or research specialist after the PICO/PECO step. The pair can discuss, refine, and complete the study plan together. At the final stage, the system assembles the approved project materials into a manuscript draft and an ethics-submission package for the relevant Institutional Ethics Committee (IEC). The application prepares materials; it does not grant ethics approval or guarantee publication.

Primary users and portals

Researcher Portal	Doctor / Research Specialist Portal
Create a project; use AI to clarify the idea; review evidence; formulate PICO/PECO; request a specialist; co-edit the plan; track tasks; prepare documents.	View collaboration requests; review the research question and evidence; chat with the researcher; advise on design, methods, sampling, analysis, and ethics documents; review drafts.

Product principles

- Human-led research: AI assists with drafting and organization; users and qualified mentors make research decisions.
- Traceability: retain sources, decisions, comments, approvals, and document versions.
- Privacy by design: minimize personal data; do not enter identifiable patient information into AI or general project notes.
- Stage-gated workflow: each stage has required inputs, review status, and a next-step action.

2. End-to-end workflow

The researcher completes the first two stages independently with the integrated AI. Once PICO/PECO is saved, the option to find and invite a doctor/research specialist becomes available. Collaboration then continues through the remaining stages to manuscript and IEC preparation.

01 · Research idea Researcher enters a clinical/research observation, topic, objective, and context. AI helps clarify the problem and propose question drafts.

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02 · Evidence + PICO/PECO AI-assisted literature discovery and evidence notes; researcher structures Population, Intervention/Exposure, Comparison, Outcome (PICO/PECO) and confirms the question.

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03 · Connect specialist Researcher selects a doctor/research specialist by specialty, expertise, language, and availability; sends a collaboration request with a project summary.

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04 · Joint discussion In-app project chat, comments, and meeting notes. Both parties refine the question, feasibility, outcomes, and scope.

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05 · Study plan Collaboratively choose study design; document setting, population, eligibility, variables, procedures, timeline, sampling, sample size, and analysis plan.

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06 · Review + protocol Specialist reviews the protocol and supporting materials; researcher addresses comments; maintain versions and decision history.

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07 · IEC readiness Generate a checklist and draft submission documents, participant information/consent materials when applicable, tools, and privacy/risk statements.

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08 · Manuscript package Compile project details, methods, results when available, references, tables, and appendices into a manuscript draft; export for human review and submission.

Flow note: The application should allow a stage to be revisited when mentor feedback changes the question, design, sample size, or protocol. Revisions should create a new version rather than silently overwrite an earlier reviewed draft.

3. Core features and functional requirements

Researcher Portal

- Account and profile: sign-up/login, research interests, department/institution (optional), language, and notification preferences.
- Project workspace: create/edit project, stage tracker, saved drafts, tasks, collaborators, files, comments, and version history.
- Integrated AI companion: guided prompts, plain-language explanations, question refinement, structured drafts, and missing-information checks. Clearly label AI-generated content and require user confirmation.
- Evidence workspace: search/discover relevant literature, save citations, summarize abstracts or supplied papers, and link claims to source records. Display limitations and avoid fabricated references.
- PICO/PECO builder: guided fields, question preview, completeness checks, and a saved checkpoint that unlocks specialist discovery.
- Specialist discovery and request: searchable doctor/research-specialist directory, expertise profile, request status, accept/decline, and project-scoped access.
- Document center: protocol, questionnaire, participant-facing materials, data dictionary, manuscript draft, export, and revision history.

Doctor / Research Specialist Portal

- Profile and expertise: specialty, research interests, experience, institution/department, languages, and collaboration preferences; verification may be added as an administrative process.
- Request inbox: view project summary and PICO/PECO, accept/decline, request clarification, or suggest another specialist.
- Collaborative review: project chat, threaded comments, inline suggestions, stage reviews, task assignment, and meeting notes.
- Research guidance: advise on design, methods, feasibility, sampling, sample size assumptions, outcomes, analysis, and ethics readiness; record rationale and unresolved questions.
- Manuscript and IEC review: comment on drafts and mark review status. Final submission and institutional approval remain with the responsible investigators and IEC.

Shared workflow and administration

- Role-based permissions: researcher owns the project; invited specialist receives only project-scoped access; revoke access when collaboration ends.
- Notifications: request updates, mentions, review requests, stage changes, and document revisions.
- Admin controls: manage accounts, specialist profiles/verification, reported content, templates, audit logs, and system configuration.
- Status model: Draft → AI-assisted → Awaiting specialist → In collaboration → Protocol review → IEC preparation → Manuscript draft → Completed/Archived.
- Exports: DOCX/PDF manuscript and protocol drafts, plus spreadsheet templates for data collection. Clearly mark draft status and version.

4. Safety, deliverables, and proposed technology stack

Ethics and responsible-use requirements

- IEC means Institutional Ethics Committee (the exact institutional name and process may vary). “IEC readiness” is a preparation checklist, not an approval, certification, or submission decision.
- The app must not provide diagnosis, treatment, or patient-specific clinical advice. It is a research-planning and collaboration tool.
- Do not upload identifiable patient data unless the institution has authorized the processing and appropriate safeguards are in place. Prefer de-identified or synthetic data during planning.
- Consent, risk/benefit, privacy, data retention, vulnerable-population safeguards, and institutional permissions must be reviewed by the investigators and relevant ethics body.
- AI outputs, literature summaries, sample-size calculations, statistical suggestions, and manuscript text require human verification. Preserve citations and flag uncertainty.
- Provide secure authentication, role-based access, encryption in transit and at rest, audit trails, backups, retention/deletion controls, and incident-response procedures.

Expected outputs

At the end of planning	At the end of the research workflow
Research question; evidence notes; PICO/PECO; specialist connection; reviewed study design and methodology; protocol draft; sampling and sample-size rationale; IEC readiness checklist.	Versioned manuscript draft; references; tables/appendices as applicable; IEC submission-preparation documents; review comments and decision history; exportable files.

Proposed technology stack (to validate during technical design)

Layer	Suggested technology
Mobile apps	React Native (TypeScript) or Flutter (Dart). Choose one for both portals unless separate native requirements justify otherwise.
Backend/API	NestJS (TypeScript) with REST APIs; WebSocket gateway for project chat, presence, and live notifications.
Database	PostgreSQL for users, projects, stages, messages, review records, citations, and audit events; Prisma or Drizzle ORM.
AI / research services	Python (FastAPI) only if needed for AI orchestration, document processing, statistics, or specialized NLP; keep model/provider integration behind a service boundary.

Search and references	PubMed / Europe PMC integrations where permitted; citation metadata storage; source URLs and retrieval timestamps.
Files and documents	S3-compatible object storage for uploads/exports; DOCX/PDF generation service; spreadsheet export for data templates.
Authentication and notifications	Secure auth with role-based authorization; push notifications via Firebase Cloud Messaging (FCM); email service for account and workflow notices.
Infrastructure and operations	Docker; cloud/VPS deployment; managed PostgreSQL or backups; HTTPS; secrets management; logging, monitoring, rate limits, and CI/CD.

Suggested MVP sequence

Phase 1: authentication, two portals, project workspace, AI-assisted idea and PICO/PECO. Phase 2: specialist directory/request, project chat, comments, and shared protocol stages. Phase 3: IEC readiness checklist, document generation, versioning, and secure exports. Phase 4: literature integrations, advanced data/statistics support, manuscript formatting, and pilot feedback.